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Subject: MDM- Wireless & Mobile Communication

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Experiment No.:	05
Title:	To measure the **Wi-Fi signal strength (RSSI)** at different distances from a Wi-Fi router and study how the signal strength changes as the distance increases.
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Marks:	
Sign of Faculty:	

Aim: To measure the Wi-Fi signal strength (RSSI) at different distances from a Wi-Fi router and study how the signal strength changes as the distance increases.

Objective: The objectives of this experiment are:

- To identify nearby Wi-Fi networks.
- To understand the concept of SSID.
- To measure Wi-Fi signal strength using RSSI.
- To observe the Wi-Fi channels used by different networks.
- To study channel congestion and interference.
- To compare signal strength at different locations.

Theory:

Basic Concept

The strength of a Wi-Fi signal is commonly represented by **RSSI (Received Signal Strength Indicator)** in **dBm**.

RSSI values are normally negative:

- **−30 dBm** → Excellent signal
- **−50 dBm** → Very good
- **−60 dBm** → Good
- **−70 dBm** → Weak but usable
- **−80 dBm** → Very weak
- **−90 dBm** → Extremely weak/no reliable connection

Remember:

A value closer to 0 dBm means a stronger signal.

For example:

−45 dBm is stronger than −70 dBm.

As the distance from the router increases, the received signal generally becomes weaker, so the RSSI becomes **more negative**.

4. Experimental Setup

Place the Wi-Fi router at a fixed location.

For example:

Wi-Fi Router

📡

|

| 1 m

|

📱 Measurement 1

|

| 2 m

|

📱 Measurement 2

|

| 5 m

|

📱 Measurement 3

|

| 10 m

|

📱 Measurement 4

|

| 15 m

|

📱 Measurement 5

Step-by-Step Procedure

Step 1: Connect Your Smartphone

On your smartphone:

1. Open **Settings**.
2. Select **Wi-Fi**.
3. Turn Wi-Fi ON.
4. Connect to the Wi-Fi network of your router.
5. Turn **Location** ON if your analyzer application requires it.

Step 2: Open Wi-Fi Analyzer

Open your Wi-Fi Analyzer application.

You should see information similar to:

- Network: MyWiFi
- Frequency: 2.4 GHz
- Channel: 6

Signal Strength: -45 dBm

The important value for this experiment is:

Signal Strength / RSSI (dBm)

Step 3: Take Measurement at 1 m

Use a measuring tape to measure exactly **1 m** from the router.

Place the smartphone at this position.

Keep the phone:

- approximately at the same height,
- facing approximately the same direction,
- stationary for a few seconds.

Record the RSSI.

For example:

- Distance = 1 m

  RSSI = −42 dBm

Step 4: Take Measurement at 2 m

Move the smartphone to **2 m** from the router.

Wait approximately **5–10 seconds** for the reading to stabilize.

Record the RSSI.

Example:

- Distance = 2 m

  RSSI = −48 dBm

Step 5: Take Measurement at 5 m

Move the smartphone to **5 m**.

Again, wait for the reading to stabilize and record the RSSI.

Example:

- Distance = 5 m

  RSSI = −57 dBm

Step 6: Take Measurement at 10 m

Move the smartphone to **10 m**.

Record the RSSI.

Example:

- Distance = 10 m

RSSI = −67 dBm

Step 7: Take Measurement at 15 m

Finally, move the smartphone to **15 m** from the router.

Record the RSSI.

Example:

- Distance = 15 m

RSSI = −75 dBm

The values above are **sample readings only**. Your actual readings will depend on the router, phone, walls, interference, antenna orientation, and environment.

6. Observation Table

You can prepare the following table in your laboratory notebook:

Sr. No.	Distance (m)	RSSI (dBm)
1	1	−42
2	2	−48
3	5	−57
4	10	−67
5	15	−75

Better experimental practice

For more reliable results, take **3 readings at each distance** and calculate their average.

Distance	Reading 1	Reading 2	Reading 3	Average RSSI
1 m	−-33	−32	−41	−42.0
2 m	−43	−42	−49	−48.0
5 m	−54	−52	−56	−57.0
10 m	−60	−59	−68	−67.0
15 m	−63	−65	−76	−75.0

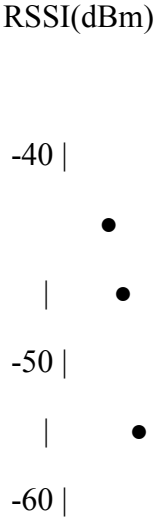
This reduces the effect of temporary fluctuations in Wi-Fi signal strength.

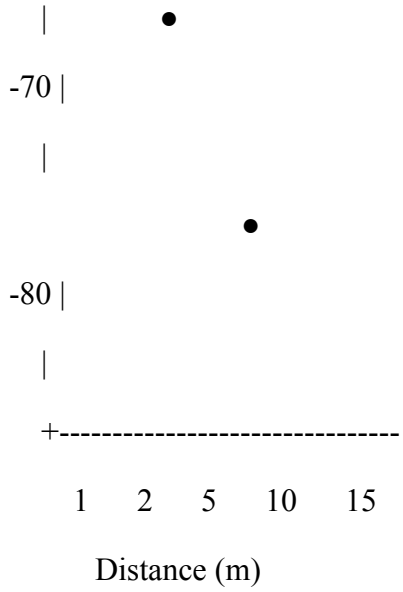
7. Plot the Graph

Plot:

- **X-axis → Distance in metres (m)**
- **Y-axis → RSSI in dBm**

For example:



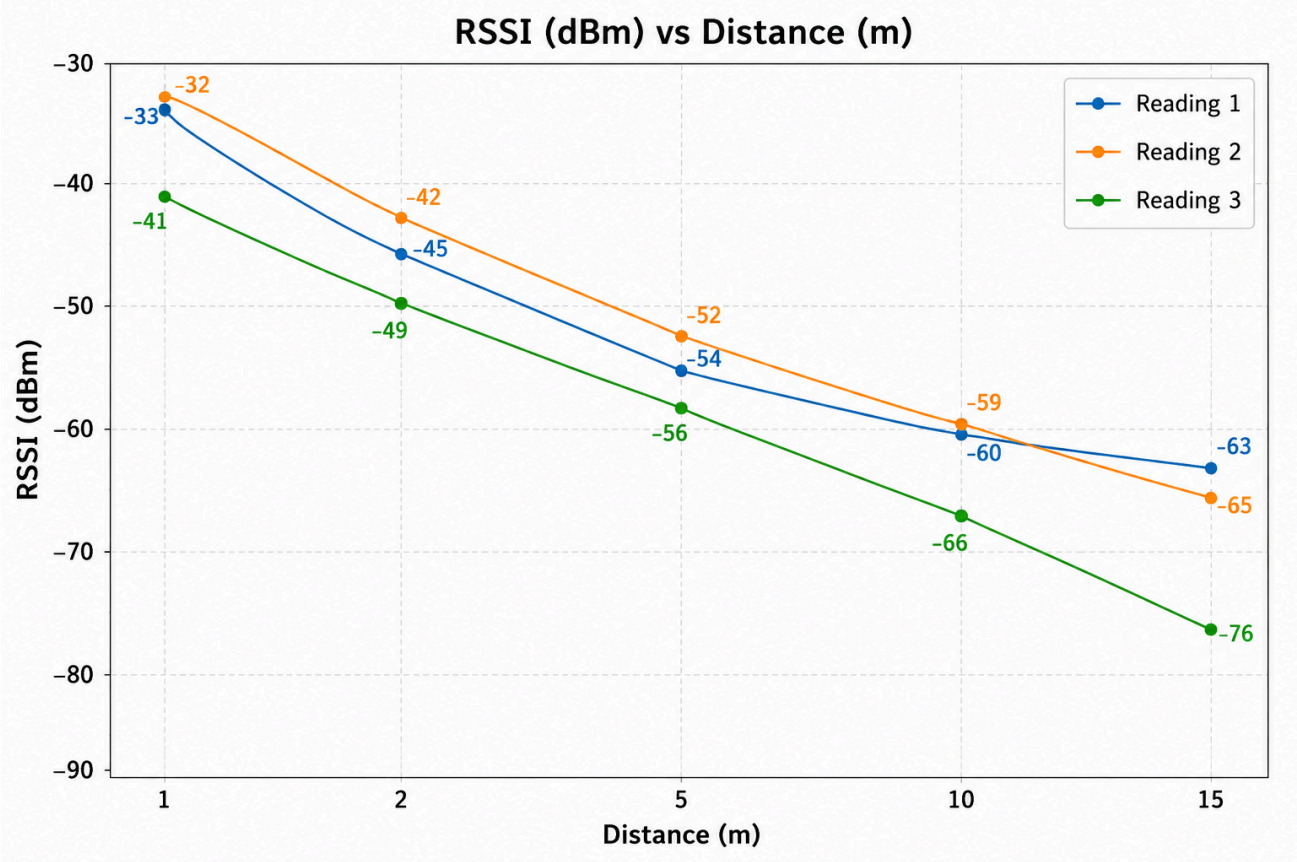


The graph should generally show that **RSSI decreases as distance increases**.

Since RSSI is negative, "decreases" means that it moves farther away from zero, for example:

−42 → −48 → −57 → −67 → −75 dBm

Code/Output:



Conclusion

The nearby Wi-Fi networks were successfully scanned using a mobile Wi-Fi analyzer application. The SSID, RSSI, operating channels, frequency bands, and channel occupancy of the detected wireless networks were recorded and analyzed. The experiment demonstrated that RSSI indicates signal strength, while channel occupancy helps identify congested and less congested channels. The practical study provided a clear understanding of how Wi-Fi networks operate and how proper channel selection contributes to efficient and reliable wireless communication.